

STP 470/Fall 2025
HW (pen-and-paper) Linear Regression

1. **(30 pts.)** Given paired data $\{(x_1, y_1), \dots, (x_n, y_n)\} \subset \mathbb{R}^2$, the **sample (Pearson) correlation coefficient**, r , is given by

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

Consider the linear regression estimates $\{(x_1, \hat{y}_1), \dots, (x_n, \hat{y}_n)\}$, where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and $\hat{\beta}_0, \hat{\beta}_1$ minimize the *residual sum of squares*, $RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$.

Prove that

$$r^2 = R^2.$$

where

$$R^2 = 1 - \frac{RSS}{TSS} \text{ and } TSS = \sum_{i=1}^n (y_i - \bar{y})^2.$$

2. **(30 pts.)** Prove that the (simple) linear regression line $y = \hat{\beta}_0 + \hat{\beta}_1 x$ passes through the point $(\bar{x}, \bar{y})$, where

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i.$$

3. **(40 pts.)** Let $d \leq n$ and assume that the matrix $X \in \mathbb{R}^{n \times d}$ is full rank, i.e., $\text{rank}(X) = d$.

- (a) Consider solving the linear system

$$X\theta = 0$$

- i. What is the dimension of the solution space $SOL(X) = \{\theta \in \mathbb{R}^d \mid X\theta = 0\}$?
- ii. What about the dimension of the solution space $SOL(X^T) = \{y \in \mathbb{R}^n \mid X^T y = 0\}$?

- (b) Consider the multiple linear regression problem with training data

$$(x_{i1}, \dots, x_{ip}, y_i), i = \overline{1, n} \quad (n > p).$$

Let $x^{(k)} = (x_{1k}, \dots, x_{nk})^T$ be the column vector containing the k -th predictor values, $k = \overline{1, p}$ and let $\mathbf{1} \in \mathbb{R}^n$, $\mathbf{1} = (1, \dots, 1)^T$. The design matrix, $X \in \mathbb{R}^{n \times (p+1)}$ is defined column-wise by

$$X = \left(\mathbf{1}, x^{(1)}, \dots, x^{(p)} \right).$$

The linear regression model provides the predictions $\hat{y}$,

$$\hat{y} = X\hat{\beta},$$

where $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)^T$ is found as the (unique) minimizer of the residual sum of squares.

i. Show that

$$RSS = (X\hat{\beta} - y)^T(X\hat{\beta} - y),$$

where y is the vector of observed values, i.e. $y = (y_1, \dots, y_n)^T$. Since $\hat{\beta}$ is the β -value that minimizes the residual sum of squares, we can write

$$\hat{\beta} = \operatorname{argmin}_{\beta} \frac{1}{2} (X\beta - y)^T(X\beta - y).$$

ii. Assume that the design matrix X is full rank. Show that $\hat{\beta}$ is the unique solution of the linear system (also known as the **normal equations**):

$$X^T X \beta = X^T y.$$